

ML Mid Sem Project Evaluation

Suicidal Sentiments Identifier

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Abstract

Social media and messaging platforms are at the center of understanding human behaviour. Our aim is to create a machine learning model that detects suicidal tendencies in communication, typically social media posts and messages, which are in a text form. The model classifies the users through speech patterns and surrounding circumstances using a classification model, exploring methods such as Support Vector Machines, Naive Bayes, etc. The method ensures a powerful tool to save lives and intervene in the health-threatening behaviors of individuals.

1. Motivation

Suicidal behaviour is a serious global problem; it takes over 700,000 lives a year, placing it as one of the leading causes of death in the world. Suicide is the fourth most common cause of death among young people between the ages of 15 and 29 years. There is great awareness about mental health in today's society, but identifying people at risk has become very challenging. The motivation for this project lies in finding an effective and scalable solution for early identification of suicidal tendencies through machine learning, which can save many lives with timely intervention.

2. Problem Statement Description

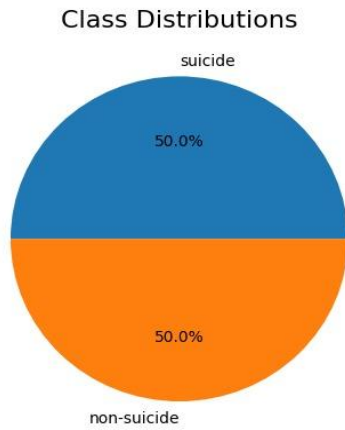
We understand the difficulty in understanding and identifying the suicidal tendency of a person based on the context of the text they post or use in conversation. With the increasing use of these platforms, people often express their emotional struggles and mental health issues in posts or messages. The challenge is to create a machine-learning model that can analyze this text and detect warning signs of suicidal thoughts by classifying users based on their speech patterns and contextual factors. The ultimate goal is to provide timely intervention by recognizing these tendencies and potentially saving lives.

3. Literature Survey

1. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9565132/>: This paper uses data of posts by individuals on reddit. It uses various DL methods and ML classifiers and conducts a binary classification to detect tweets that may have suicidal tendencies.
2. https://www.researchgate.net/publication/323065338_Online_suicide_prevention_through_optimised_text_classification: This paper investigates a method to automatically classify suicide-related Dutch forum posts. They used Naive Bayes and Support Vector Machine for classification.
3. https://www.researchgate.net/publication/259948193_Predicting_Depression_via_Social_Media: This paper explores social media and investigates Major Depressive Disorder among Twitter users.

4. Dataset Details

The dataset used for this project is a publicly available Reddit dataset sourced from Kaggle, containing approximately 230,000 posts from the **r/SuicideWatch** subreddit. These posts span a period from **December 2008 to January 2021**, covering a wide range of discussions related to suicidal tendencies and mental health support. Non-suicide posts are collected from **r/teenagers**. The dataset is evenly split between posts categorized as suicidal and non-suicidal. A key aspect of the dataset is the variation in post lengths. The average post length is **131.5 words**, with the shortest post consisting of just one word and the longest post reaching a maximum of 15,632 words. Most posts are relatively short, with a large portion (80%) containing fewer than 190 words. The length distribution is skewed, with fewer posts having longer lengths.



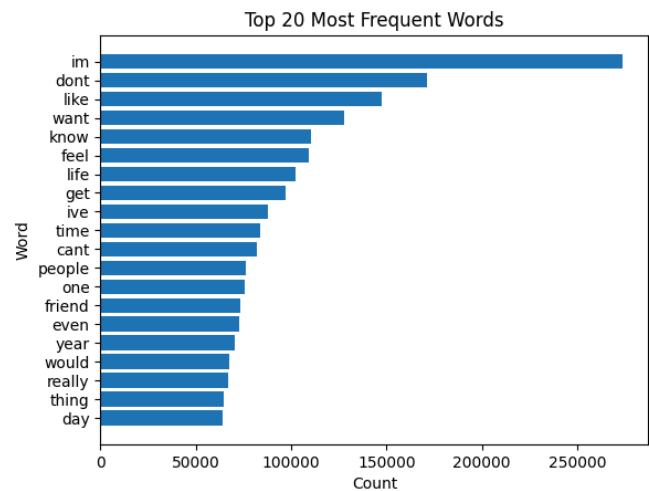
5. Data Pre Processing

A comprehensive data preprocessing was implemented in multiple stages to clean and prepare the text data for accurate analysis and modeling. The first step involved converting all text to lowercase, ensuring uniformity by eliminating case sensitivity so that words like "Suicide" and "suicide" are treated equivalently. This was followed by the removal of URLs, as links provide no meaningful contribution to the analysis of suicidal tendencies and can introduce unnecessary noise into the data. Numerical values, including phone numbers and underscores, were stripped from the text to avoid irrelevant information, simplifying the data for easier processing. Punctuation and special characters were also removed to clean the text further and focus solely on the meaningful content. The text was then tokenized, breaking down the posts into individual words and facilitating more granular analysis. Common stopwords (such as "the," "is," and "for") that do not contribute significantly to the meaning or context of the posts were removed to emphasize more relevant words. **Lemmatization** was applied to reduce words to their root or base form, effectively handling variations in tense and conjugation. For instance, words like "running," "ran," and "runs" were reduced to their root form "run." This ensures that all word variations are analysed uniformly, providing better consistency and improving the performance of machine learning models.

6. Methodology: Feature Extraction

For feature extraction, multiple techniques were selected to optimise the identification and working of the model. The primary goal of feature extraction was to simplify the data by highlighting significant terms, thereby improving model performance and interpretability. This process helps reduce noise in

the dataset and allows the model to focus on the most relevant patterns associated with suicidal thoughts. Deploying **Bag of Words (BoW)** and **Term Frequency-Inverse Document Frequency (TF-IDF)** methods for feature selection. Both techniques are widely recognised for effectively capturing important features for sentiment analysis in text data. BoW represents text based on word frequency, while TF-IDF enhances this by considering the relative importance of words across multiple posts and assigning them appropriate weights. These approaches ensure that significant terms are prioritised, improving the model's ability to detect patterns accurately. The number of features depends on the size of the vocabulary, which consists of all the unique words in the entire dataset (164,765 in our case).



7. Methodology: Model Selection

Since the task at hand is a binary classification problem of labelling the text as Suicidal or Non-Suicidal, we explored multiple classification algorithms to determine the best fit for this specific use case. We initially chose Logistic Regression because of its interpretability, as it provides clear coefficients indicating how each word influences the prediction of suicidal tendencies. Its straightforward nature makes it ideal for gaining insights into the model's decision-making process. We also explored Multinomial Naive Bayes because of its probabilistic nature in handling the word frequencies, as it was well suited for quick categorisation. Still, we resulted in less accuracy as it is Naive. Perceptron, on the other hand, is an iterative learning algorithm. Still, as soon as it converges, the algorithm stops and does not optimise the hyperplane further, which results in less accuracy.

8. Results and Analysis

To evaluate the models implemented, we use the Accuracy, Recall and F1 scores of the models. Accuracy tells how many predictions are actually correct out of all positive/negative predictions. Recall tells us how many of all positive/negative classes we could predict. F1 score provides an outlook of classification for each case (i.e. suicidal and non-suicidal). For metrics, we use the values (TP, TN, FP, FN) obtained from the confusion matrix which are described as follows:

$$Accuracy = \frac{TP + TN}{TP + TN + FN + FP}$$

$$Precision = \frac{TP}{TP + FP}$$

$$Recall = \frac{TP}{TP + FN}$$

$$F1 = 2 \times \frac{(Precision * Recall)}{(Precision + Recall)}$$

Accuracy: 0.9196				
Classification Report:				
	precision	recall	f1-score	support
non-suicide	0.90	0.94	0.92	23287
suicide	0.94	0.90	0.92	23128
accuracy			0.92	46415
macro avg	0.92	0.92	0.92	46415
weighted avg	0.92	0.92	0.92	46415
[[21876 1411]				
[2321 20807]]				

Table 1: Performance on Perceptron Model

Accuracy: 0.8885				
Classification Report:				
	precision	recall	f1-score	support
non-suicide	0.97	0.80	0.88	23287
suicide	0.83	0.98	0.90	23128
accuracy			0.89	46415
macro avg	0.90	0.89	0.89	46415
weighted avg	0.90	0.89	0.89	46415
[[18672 4615]				
[559 22569]]				

Table 2: Performance on Multinomial Naive Bayes Model

Accuracy: 0.9364				
Classification Report:				
	precision	recall	f1-score	support
non-suicide	0.93	0.94	0.94	23287
suicide	0.94	0.93	0.94	23128
accuracy			0.94	46415
macro avg	0.94	0.94	0.94	46415
weighted avg	0.94	0.94	0.94	46415
[[21988 1299]				
[1652 21476]]				

Table 3: Performance of Logistic Regression Model

Each of the tables above show the difference between the performance of each model. From Table 1, we can infer that the performance offered by the Perceptron model (accuracy = 91.96%, f1 score = 92%) lies in the middle ground of the other two models. Table 2 shows that the Multinomial Naive Bayes model has the worst performance out of all the models that were tested (accuracy = 88.85%, f1 score = 88% for the non-suicide class and 90% for the suicide class) and has the highest difference in non-suicide recall and suicide recall. It also has a huge difference in the precision percentage of suicide calls and non-suicide calls which is not good as it cannot classify the non-suicide texts as accurately as the suicide texts which can lead to more false positive alerts. Table 3 shows the performance of the Logistic Regression Model, which through comparison proves to be the best-performing model achieving an accuracy of 93.64% and an F1 score of 94%. Hence, we chose to proceed with our Logistic Regression Model.

9. Conclusion

Many people use social media as a platform to convey their feelings and troubles they are going through, or sometimes explicit suicide notes. With the increasing suicide rates especially in India, the earlier detection of suicide intent, especially among the youth can save lives. Our proposed model would be useful in detecting the user with such intent. Our model performs reasonably well, achieving a maximum accuracy of 93.64%. We have implemented some of the best practices from the domain of NLP using classical Machine Learning.

However, there is still a lot of room for improvement. We still need to do hyperparameter tuning and test various other classifiers. We also need to work around regularization. We have completed work until Week 10 and are ahead of schedule. However, we would like to implement

certain improvements in our model, particularly regarding NLP. We aim to explore using large language models (LLMs) to correct spelling mistakes in our dataset, enhancing its data quality. Additionally, we plan to experiment with more complex models such as Multi-Layer Perceptrons (MLP) and Random Forests.

Each team member contributed equally to the project, with distinct responsibilities. Himanshu worked on the methodology and feature selection and extraction. Dhruv handled the ML model training and selection and the literature review. Sarthak was responsible for the literature review, dataset preprocessing, and result analysis. Aayush focused on ML model selection, comparison, inference, and analysis.

References

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